LUMINAIRE PHOTOMETRIC TEST REPORT ของหลอดไฟ A2

LUMINOUS DISTRIBUTION INTENSITY DATA

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

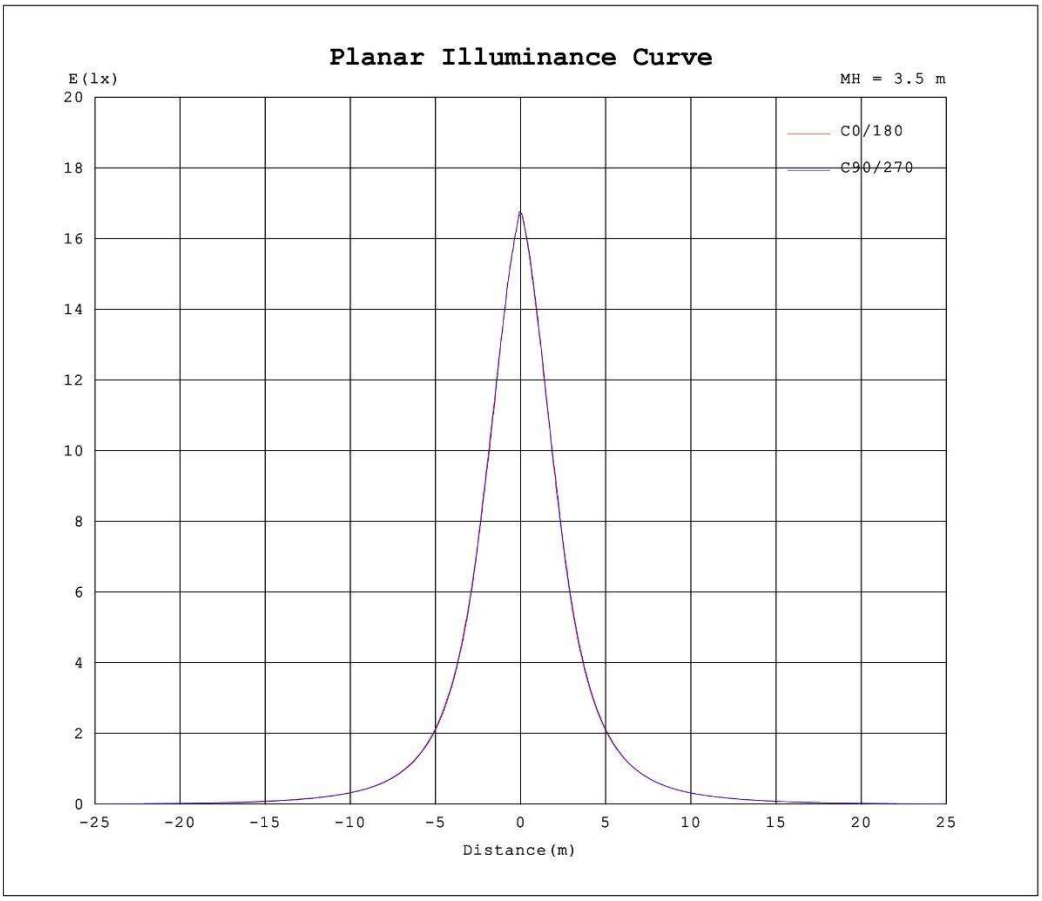
Table--1 UNIT: cd

γ (DEG) \ C (DEG)	0	15	30	45	60	75	90	105	120	135	150	165	180	195	210	225	240	255	270
0	204	206	205	205	206	206	205	206	205	206	205	206	204	206	205	205	206	206	205
5	199	200	199	199	199	199	200	200	199	200	199	200	199	200	200	200	200	201	200
10	196	195	195	195	196	195	195	194	194	195	194	195	195	195	194	194	196	195	196
15	191	191	191	190	190	190	189	190	188	188	188	190	189	190	190	189	190	190	191
20	186	187	187	185	185	185	185	185	184	185	184	185	185	186	186	186	186	186	186
25	182	182	182	181	181	180	180	180	179	180	179	180	180	180	181	181	181	181	182
30	177	178	177	177	176	175	175	174	174	175	174	175	175	176	176	176	176	177	177
35	171	172	170	170	170	169	169	168	167	167	168	168	168	168	168	168	170	169	170
40	163	164	162	162	161	160	161	160	159	159	159	159	159	159	161	159	160	161	162
45	155	155	155	155	154	153	152	152	151	150	150	151	151	151	152	152	152	152	153
50	147	147	147	146	146	145	145	143	143	143	142	143	143	143	144	143	144	145	145
55	140	140	140	139	139	138	137	136	135	136	135	136	136	137	136	136	137	137	137
60	132	133	132	132	132	131	130	129	128	129	128	129	128	129	129	128	129	129	130
65	124	125	124	124	123	123	122	121	120	120	120	120	120	120	120	120	121	121	121
70	115	116	115	115	114	113	113	112	111	111	110	111	110	111	111	110	112	111	112
75	104	104	103	102	101	99.7	97.1	97.7	96.2	97.3	95.9	97.8	97.5	98.3	98.4	98.5	99.4	99.9	101
80	76.2	76.3	74.2	74.0	72.0	71.5	67.0	70.2	68.7	69.7	68.8	70.4	70.3	70.4	69.9	69.6	72.7	73.8	75.5
85	43.1	43.7	42.0	42.1	40.5	40.8	36.2	39.5	38.7	39.6	38.8	40.2	39.8	39.5	37.9	38.3	40.6	41.6	43.3
90	15.5	15.6	14.9	15.1	14.6	14.6	10.6	14.3	14.0	14.4	14.1	14.5	14.2	14.1	11.3	13.3	13.5	14.6	15.3
95	0.27	0.23	0.27	0.23	0.24	0.17	0.24	0.20	0.24	0.20	0.27	0.20	0.27	0.30	0.23	0.30	0.24	0.23	0.27
100	0.23	0.30	0.23	0.24	0.13	0.27	0.20	0.27	0.23	0.30	0.23	0.27	0.30	0.27	0.30	0.27	0.34	0.23	0.30
105	0.23	0.30	0.23	0.33	0.13	0.27	0.20	0.27	0.23	0.34	0.23	0.37	0.43	0.27	0.44	0.27	0.37	0.23	0.37
110	0.23	0.37	0.23	0.33	0.13	0.30	0.20	0.30	0.23	0.40	0.23	0.44	0.43	0.27	0.44	0.27	0.40	0.23	0.40
115	0.23	0.37	0.23	0.40	0.13	0.37	0.20	0.34	0.23	0.40	0.23	0.44	0.43	0.27	0.47	0.27	0.44	0.23	0.44
120	0.23	0.40	0.23	0.40	0.13	0.44	0.20	0.37	0.23	0.44	0.23	0.44	0.47	0.27	0.50	0.27	0.47	0.23	0.47
125	0.23	0.44	0.23	0.43	0.13	0.44	0.20	0.44	0.23	0.50	0.23	0.47	0.50	0.27	0.50	0.27	0.50	0.23	0.50
130	0.23	0.47	0.23	0.47	0.20	0.50	0.20	0.47	0.23	0.47	0.23	0.47	0.50	0.27	0.57	0.27	0.50	0.23	0.50
135	0.23	0.50	0.23	0.50	0.20	0.50	0.20	0.50	0.23	0.50	0.23	0.47	0.50	0.27	0.57	0.27	0.50	0.23	0.50
140	0.23	0.50	0.23	0.53	0.34	0.50	0.20	0.54	0.23	0.50	0.23	0.47	0.50	0.27	0.57	0.27	0.50	0.23	0.50
145	0.23	0.54	0.27	0.53	0.37	0.54	0.20	0.57	0.23	0.50	0.23	0.47	0.50	0.34	0.57	0.27	0.50	0.23	0.50
150	0.23	0.57	0.27	0.53	0.44	0.54	0.23	0.57	0.23	0.50	0.27	0.47	0.50	0.34	0.57	0.37	0.50	0.30	0.50
155	0.30	0.57	0.30	0.53	0.44	0.54	0.30	0.57	0.30	0.50	0.37	0.47	0.50	0.44	0.57	0.40	0.50	0.37	0.50
160	0.43	0.57	0.40	0.53	0.44	0.54	0.40	0.57	0.34	0.50	0.40	0.47	0.50	0.47	0.57	0.47	0.50	0.44	0.50
165	0.43	0.57	0.47	0.53	0.47	0.54	0.44	0.57	0.40	0.50	0.44	0.47	0.47	0.47	0.57	0.43	0.50	0.44	0.50
170	0.43	0.57	0.47	0.53	0.47	0.54	0.44	0.57	0.44	0.50	0.44	0.47	0.47	0.47	0.57	0.43	0.50	0.44	0.50
175	0.47	0.57	0.47	0.53	0.50	0.54	0.47	0.57	0.47	0.50	0.47	0.47	0.47	0.47	0.60	0.47	0.50	0.50	0.50
180	0.47	0.57	0.50	0.50	0.54	0.54	0.47	0.57	0.47	0.47	0.47	0.47	0.47	0.47	0.47	0.50	0.50	0.50	0.50

C Range: 0 - 360DEG
C Interval: 15.0DEG
Test Speed: HIGH
Temperature:26.8DEG
Operators:Kitivut
Test Date:06 June 2025

γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

Planar Illuminance Curve



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LED Avg.L Report

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

AvgL	cd/m2
L_0~180(65)av	5880
L_0~180(75)av	7917
L_0~180(85)av	9689
L_90~270(65)av	5851
L_90~270(75)av	7804
L_90~270(85)av	9296
L_45(65)av	5853
L_45(75)av	7873
L_45(85)av	9537

Standard: GB/T 29293-2012

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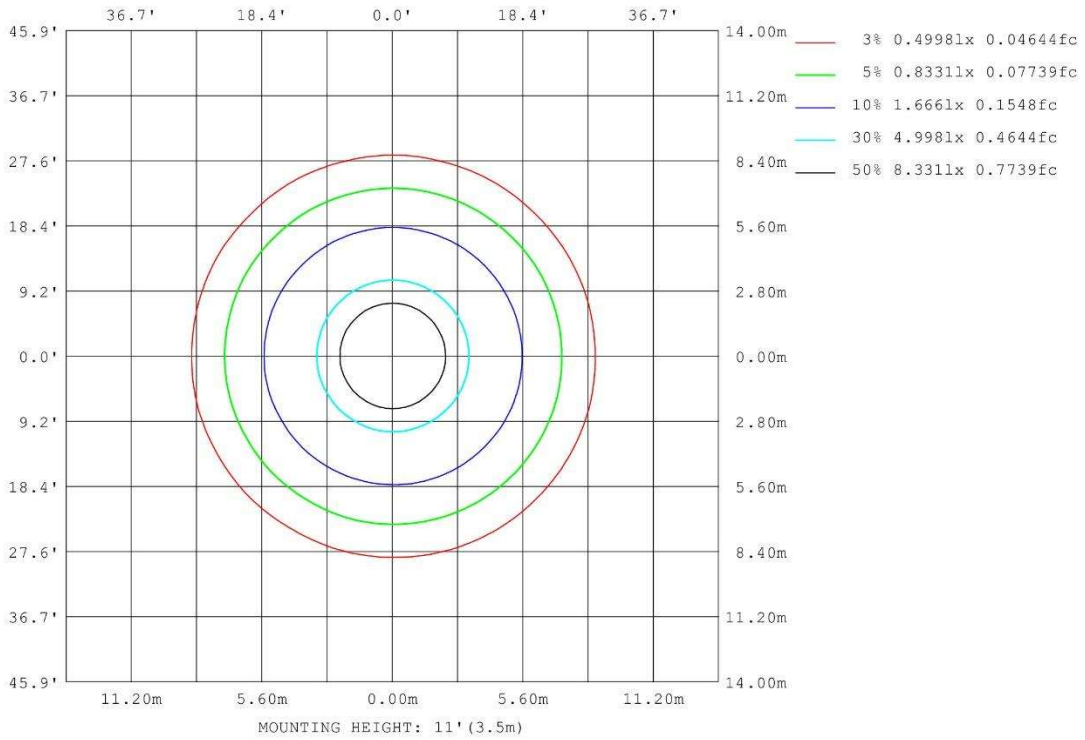
ISOLUX DIAGRAM

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:



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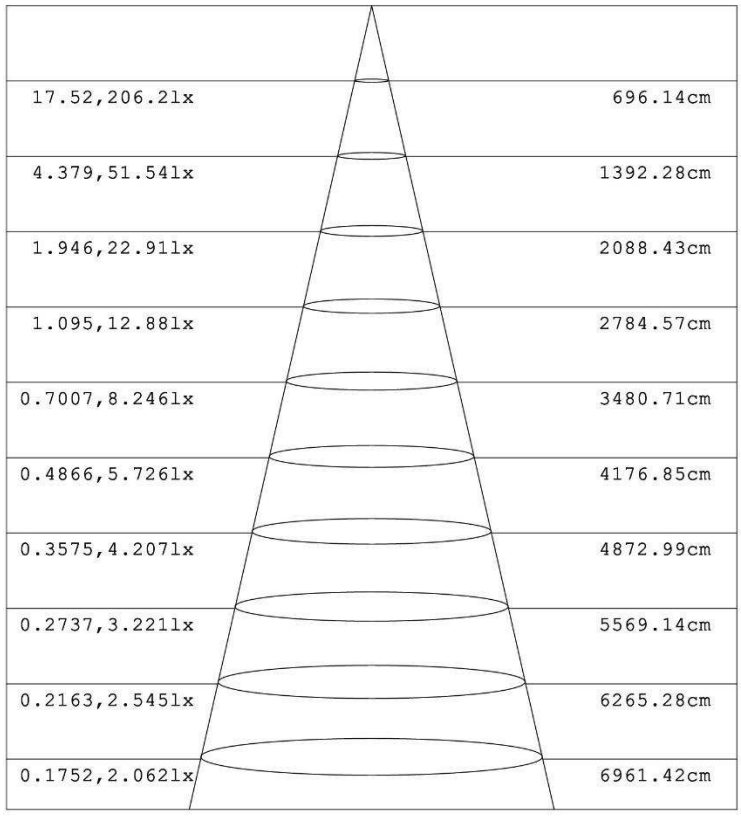
AAI Figure

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

Flux out:669.3 lm



Height	Eavg,Emax	Diameter
1m	17.52,206.2lx	696.14cm
2m	4.379,51.54lx	1392.28cm
3m	1.946,22.91lx	2088.43cm
4m	1.095,12.88lx	2784.57cm
5m	0.7007,8.246lx	3480.71cm
6m	0.4866,5.726lx	4176.85cm
7m	0.3575,4.207lx	4872.99cm
8m	0.2737,3.221lx	5569.14cm
9m	0.2163,2.545lx	6265.28cm
10m	0.1752,2.062lx	6961.42cm

Angle:147.94deg

Note:The Curves indicate the illuminated area and the average illumination when the luminaire is at different distance.

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γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
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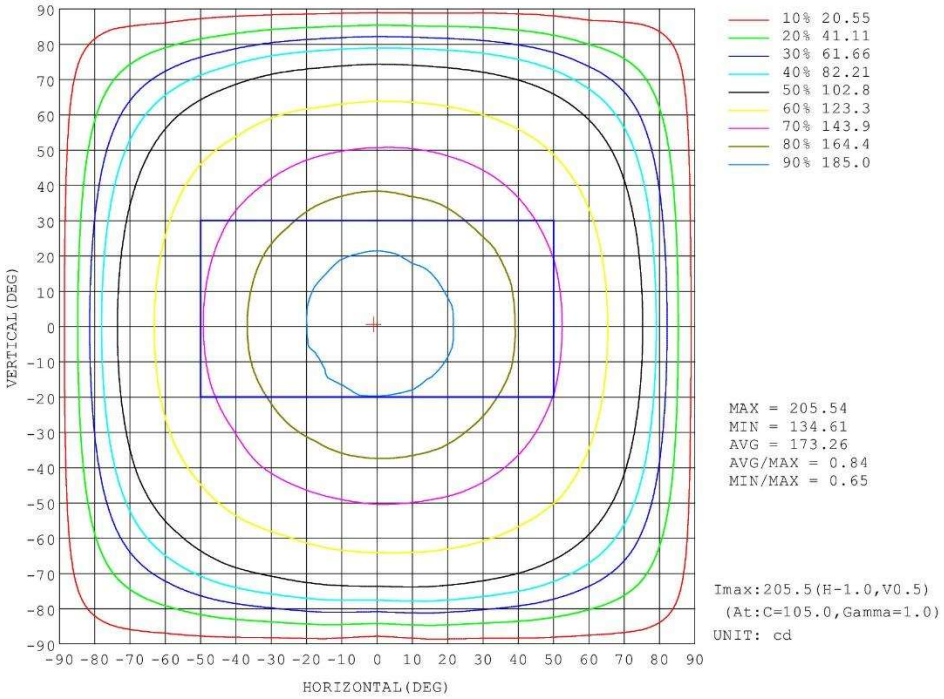
ISOCANDELA DIAGRAM

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:



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Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

UTILIZATION FACTORS TABLE

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

REFLECTANCE										
Ceiling	0.8	0.8	0.8	0.7	0.7	0.7	0.5	0.5	0.5	0
Walls	0.7	0.5	0.3	0.7	0.5	0.3	0.7	0.5	0.3	0
Working plane	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0
ROOM INDEX	UTILIZATION FACTORS(PERCENT) k(RI) x RCR = 5									
k = 0.60	52	39	32	51	39	32	50	38	31	24
0.80	61	48	40	60	48	40	58	47	40	32
1.00	69	57	48	68	56	48	65	58	47	39
1.25	76	64	56	75	63	55	72	62	54	46
1.50	82	70	61	80	69	61	76	67	60	51
2.00	89	79	71	87	77	70	83	75	68	59
2.50	94	84	77	91	83	76	87	80	74	64
3.00	97	89	82	95	87	81	91	84	79	69
4.00	102	95	89	100	93	88	95	90	85	75
5.00	105	99	94	103	97	92	98	93	89	79
ROOM INDEX	UF(total)									Direct
According to DIN EN 13032-2 2004				Suspended					SHRNOM = 1.25	

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γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
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Remarks: D3 with intensity Cal
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UGR(Unified Glare Rating) Table

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

ceiling/cavity	0.7	0.7	0.5	0.5	0.3	0.7	0.7	0.5	0.5	0.3
walls	0.5	0.3	0.5	0.3	0.3	0.5	0.3	0.5	0.3	0.3
working plane	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Room dimensions	Viewed crosswise					Viewed endwise				
x = 2H y = 2H	18.7	20.3	18.9	20.6	20.8	18.6	20.2	18.9	20.5	20.7
3H	21.3	22.9	21.6	23.1	23.4	21.2	22.8	21.5	23.0	23.3
4H	22.7	24.2	23.0	24.5	24.8	22.5	24.0	22.9	24.3	24.6
6H	23.9	25.3	24.3	25.6	25.9	23.6	25.0	24.0	25.3	25.6
8H	24.4	25.8	24.7	26.1	26.4	24.0	25.4	24.4	25.7	26.0
12H	24.8	26.1	25.1	26.4	26.7	24.3	25.7	24.7	26.0	26.3
4H 2H	19.6	21.1	19.9	21.3	21.6	19.5	21.0	19.9	21.3	21.6
3H	22.4	23.8	22.8	24.1	24.4	22.3	23.7	22.7	24.0	24.3
4H	24.0	25.2	24.4	25.5	25.9	23.8	25.1	24.2	25.4	25.8
6H	25.3	26.4	25.7	26.8	27.2	25.1	26.2	25.5	26.5	26.9
8H	25.9	26.9	26.3	27.3	27.7	25.5	26.6	26.0	27.0	27.4
12H	26.3	27.3	26.8	27.7	28.1	25.9	26.9	26.4	27.3	27.7
8H 4H	24.5	25.6	25.0	26.0	26.4	24.4	25.5	24.9	25.9	26.3
6H	26.1	27.0	26.6	27.4	27.8	25.9	26.7	26.3	27.2	27.6
8H	26.8	27.5	27.2	28.0	28.5	26.5	27.3	26.9	27.7	28.2
12H	27.4	28.0	27.8	28.5	29.0	27.0	27.7	27.5	28.1	28.6
12H 4H	24.6	25.6	25.1	26.0	26.4	24.5	25.5	25.0	25.9	26.3
6H	26.2	27.0	26.7	27.5	27.9	26.0	26.8	26.5	27.3	27.7
8H	27.0	27.7	27.5	28.1	28.6	26.7	27.4	27.2	27.9	28.4
Variations with the observer position at spacings(CIE Pub.117):										
S = 1.0H	+ 0.1 / - 0.1					+ 0.1 / - 0.1				
1.5H	+ 0.1 / - 0.2					+ 0.2 / - 0.2				
2.0H	+ 0.2 / - 0.4					+ 0.2 / - 0.4				

CIE Pub.117, 776.9 lm Total Lamp Luminous Flux Corrected (8log(F/F0) = -0.9)
Area: 0.0490874 m2

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Humidity:42.9%
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WEC AND CCEC

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
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ρcc	80%			70%			50%			30%			10%			0
ρw	50%	30%	10%	50%	30%	10%	50%	30%	10%	50%	30%	10%	50%	30%	10%	0
ρfc	20%			20%			20%			20%			20%			0
RCR	RCR:Room Cavity Ratio						Wall Exitance Coeffcients(WEC)									
0.0																
1.0	.393	.223	.071	.385	.220	.070	.371	.213	.068	.358	.207	.066	.346	.201	.065	
2.0	.349	.191	.059	.342	.188	.058	.330	.183	.057	.318	.178	.056	.307	.174	.055	
3.0	.312	.166	.050	.306	.164	.049	.295	.160	.049	.284	.156	.048	.275	.152	.047	
4.0	.281	.146	.043	.276	.144	.043	.266	.141	.042	.256	.138	.042	.248	.135	.041	
5.0	.255	.130	.038	.250	.128	.037	.241	.126	.037	.233	.123	.037	.226	.121	.036	
6.0	.233	.117	.033	.229	.116	.033	.221	.113	.033	.214	.111	.033	.207	.109	.032	
7.0	.214	.106	.030	.211	.105	.030	.204	.103	.030	.197	.101	.029	.191	.099	.029	
8.0	.199	.097	.027	.195	.096	.027	.189	.094	.027	.183	.093	.027	.178	.091	.027	
9.0	.185	.089	.025	.182	.089	.025	.176	.087	.025	.171	.086	.025	.166	.084	.024	
10.0	.173	.083	.023	.170	.082	.023	.165	.081	.023	.160	.080	.023	.156	.078	.022	

ρcc	80%			70%			50%			30%			10%			0
ρw	50%	30%	10%	50%	30%	10%	50%	30%	10%	50%	30%	10%	50%	30%	10%	0
ρfc	20%			20%			20%			20%			20%			0
RCR	RCR:Room Cavity Ratio						Ceiling Cavity Exitance Coefficients(CCEC)									
0.0	.195	.195	.195	.167	.167	.167	.114	.114	.114	.065	.065	.065	.021	.021	.021	
1.0	.191	.160	.133	.163	.137	.114	.112	.095	.079	.064	.055	.046	.021	.018	.015	
2.0	.185	.136	.096	.158	.117	.083	.108	.081	.058	.063	.047	.034	.020	.015	.011	
3.0	.177	.119	.073	.152	.103	.063	.104	.072	.045	.060	.042	.026	.019	.014	.009	
4.0	.170	.107	.058	.145	.092	.051	.100	.064	.036	.058	.038	.021	.019	.012	.007	
5.0	.162	.096	.048	.139	.083	.042	.096	.058	.030	.055	.034	.018	.018	.011	.006	
6.0	.154	.088	.041	.132	.076	.036	.091	.053	.025	.053	.031	.015	.017	.010	.005	
7.0	.146	.081	.036	.126	.071	.031	.087	.049	.022	.050	.029	.013	.016	.010	.004	
8.0	.139	.076	.032	.120	.066	.028	.083	.046	.020	.048	.027	.012	.016	.009	.004	
9.0	.133	.071	.029	.114	.061	.025	.079	.043	.018	.046	.025	.011	.015	.008	.004	
10.0	.126	.066	.026	.109	.058	.023	.076	.041	.016	.044	.024	.010	.014	.008	.003	

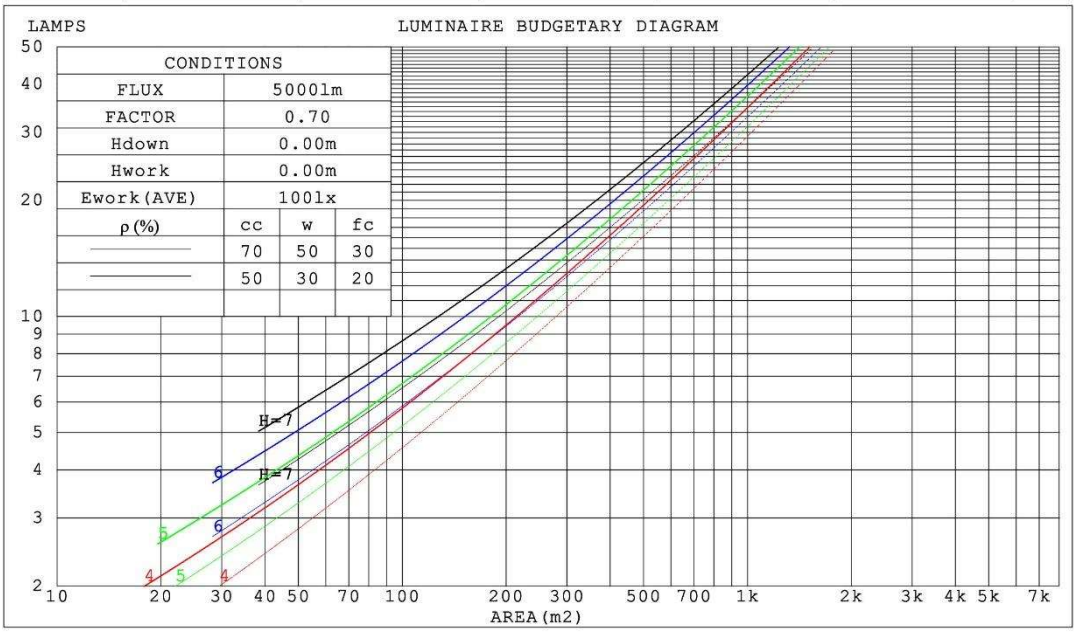
C Range: 0 - 360DEG
C Interval: 15.0DEG
Test Speed: HIGH
Temperature:26.8DEG
Operators:Kitivut
Test Date:06 June 2025

γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

CU AND LUMINAIRE BUDGETARY ESTIMATE DIAGRAM

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

ρcc	80%			70%			50%			30%			10%			0
ρw	50%	30%	10%	50%	30%	10%	50%	30%	10%	50%	30%	10%	50%	30%	10%	0
ρfc	20%			20%			20%			20%			20%			0
RCR	RCR:Room Cavity Ratio						Coefficients of Utilization(CU)									
0.0	1.19	1.19	1.19	1.16	1.16	1.16	1.11	1.11	1.11	1.06	1.06	1.06	1.01	1.01	1.01	.99
1.0	.99	.94	.89	.97	.92	.87	.92	.88	.85	.88	.85	.82	.85	.82	.79	.77
2.0	.84	.76	.69	.82	.75	.68	.78	.72	.67	.75	.70	.65	.72	.67	.63	.61
3.0	.73	.63	.56	.71	.62	.55	.68	.60	.54	.65	.58	.53	.62	.56	.52	.49
4.0	.64	.54	.46	.62	.53	.46	.59	.51	.45	.57	.50	.44	.54	.48	.43	.41
5.0	.56	.46	.39	.55	.46	.39	.53	.44	.38	.51	.43	.38	.49	.42	.37	.35
6.0	.50	.40	.33	.49	.40	.33	.47	.39	.33	.45	.38	.32	.44	.37	.32	.30
7.0	.45	.36	.29	.44	.35	.29	.43	.35	.29	.41	.34	.28	.40	.33	.28	.26
8.0	.41	.32	.26	.40	.32	.26	.39	.31	.25	.38	.30	.25	.36	.30	.25	.23
9.0	.38	.29	.23	.37	.29	.23	.36	.28	.23	.35	.28	.23	.34	.27	.22	.20
10.0	.35	.26	.21	.34	.26	.21	.33	.26	.21	.32	.25	.20	.31	.25	.20	.18



C Range: 0 - 360DEG
C Interval: 15.0DEG
Test Speed: HIGH
Temperature:26.8DEG
Operators:Kitivut
Test Date:06 June 2025

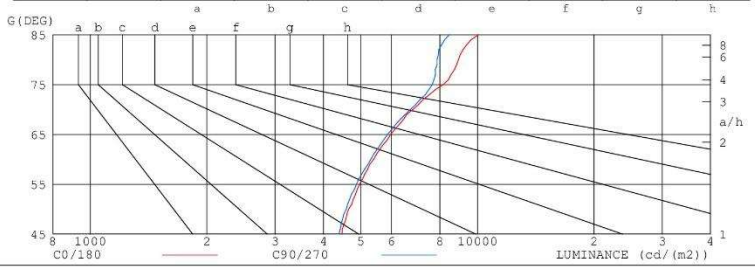
γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

LUMINANCE LIMITATION CURVES

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

LUMINANCE LIMITATION CURVES

GLARE	CLASS	ILLUMINANCE (lx)							
1.15	A	2000	1000	500	<=300				
1.50	B		2000	1000	500	<=300			
1.85	C			2000	1000	500	<=300		
2.20	D				2000	1000	500	<=300	
2.55	E					2000	1000	500	<=300
		a	b	c	d	e	f	g	h



LUMINANCE cd/(m2)		
G(DEG)	C0/180	C90/270
85	10074	8471
80	8935	7861
75	8163	7642
70	6822	6714
65	5969	5866
60	5393	5308
55	4961	4872
50	4650	4586
45	4467	4389

C Range: 0 - 360DEG
C Interval: 15.0DEG
Test Speed: HIGH
Temperature:26.8DEG
Operators:Kitivut
Test Date:06 June 2025

γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

ZONAL FLUX DIAGRAM

ZONAL FLUX DIAGRAM:

γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	Φ zone	Φ total	%lum,lamp
10	195.5	194.6	195.1	194.7	194.6	194.1	195.5	195.5	0- 10	18.93	18.93	2.44,2.44
20	186.1	185.1	184.6	185.2	184.6	185.9	186.3	187.1	10- 20	53.77	72.70	9.36,9.36
30	177.4	176.7	175.2	174.9	175.0	175.8	176.9	178.2	20- 30	83.55	156.2	20.1,20.1
40	162.8	162.1	160.7	158.5	158.8	159.5	161.6	162.3	30- 40	106.0	262.2	33.8,33.8
50	146.7	146.3	144.7	143.3	142.6	143.5	144.8	147.4	40- 50	118.2	380.4	49,49
60	132.4	131.9	130.3	128.9	128.2	128.4	129.9	131.0	50- 60	123.3	503.6	64.8,64.8
70	114.5	114.7	112.7	111.3	110.4	110.4	111.6	112.7	60- 70	120.5	624.2	80.3,80.3
80	76.16	73.98	67.01	69.74	70.28	69.63	75.49	76.09	70- 80	102.5	726.7	93.5,93.5
90	15.55	15.06	10.55	14.40	14.22	13.31	15.28	15.09	80- 90	45.34	772.0	99.4,99.4
100	0.2339	0.2362	0.2014	0.3021	0.3016	0.2675	0.3021	0.2685	90-100	2.972	775.0	99.8,99.8
110	0.2339	0.3343	0.2014	0.4028	0.4343	0.2675	0.4028	0.2685	100-110	0.3033	775.3	99.8,99.8
120	0.2339	0.4012	0.2014	0.4364	0.4678	0.2675	0.4700	0.2685	110-120	0.3197	775.6	99.8,99.8
130	0.2339	0.4672	0.2014	0.4706	0.5012	0.2675	0.5035	0.2685	120-130	0.3146	776.0	99.9,99.9
140	0.2339	0.5349	0.2014	0.5035	0.5012	0.2675	0.5035	0.2685	130-140	0.2899	776.2	99.9,99.9
150	0.2339	0.5349	0.2350	0.5035	0.5012	0.3686	0.5035	0.3021	140-150	0.2519	776.5	100,100
160	0.4338	0.5349	0.4028	0.5035	0.5008	0.4678	0.5035	0.4691	150-160	0.2058	776.7	100,100
170	0.4343	0.5349	0.4364	0.5035	0.4678	0.4346	0.5035	0.4364	160-170	0.1358	776.8	100,100
180	0.4678	0.5015	0.4700	0.4700	0.4678	0.5015	0.5035	0.4700	170-180	0.0468	776.9	100,100
DEG	LUMINOUS INTENSITY:cd									UNIT:lm		

C Range: 0 - 360DEG
C Interval: 15.0DEG
Test Speed: HIGH
Temperature:26.8DEG
Operators:Kitivut
Test Date:06 June 2025

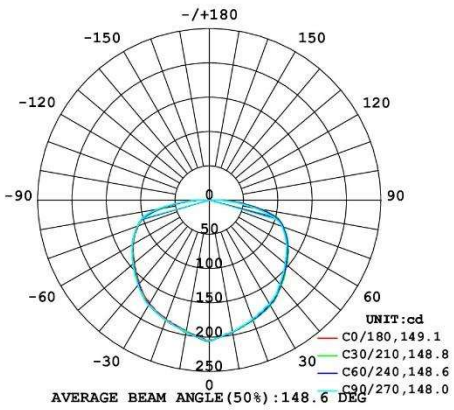
γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

LUMINAIRE PHOTOMETRIC TEST REPORT

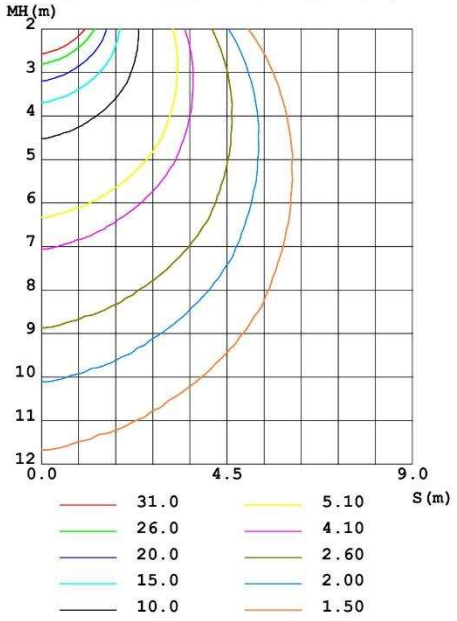
Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

DATA OF LAMP		PHOTOMETRIC DATA Eff: 98.52 lm/W			
MODEL	Bulb 6500K	Imax(cd)	206.2	S/MH(C0/180)	1.28
NOMINAL POWER(W)	8	LOR(%)	100.0	S/MH(C90/270)	1.30
RATED VOLTAGE(V)	220	TOTAL FLUX(lm)	776.89	η UP,DN(C0-180)	0.3,49.6
NOMINAL FLUX(lm)	776.887	CIE CLASS	DIRECT	η UP,DN(C180-360)	0.3,49.7
LAMPS INSIDE	1	η up(%)	0.6	CIBSE SHR NOM	1.50
TEST VOLTAGE(V)	220	η down(%)	99.4	CIBSE SHR MAX	1.55

LUMINOUS INTENSITY DISTRIBUTION DIAGRAM



C0 PLANE ISOLUX DIAGRAM (UNIT:lx)



C Range: 0 - 360DEG
C Interval: 15.0DEG
Test Speed: HIGH
Temperature:26.8DEG
Operators:Kitivut
Test Date:06 June 2025

γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

LUMINOUS DISTRIBUTION INTENSITY DATA

Test:U:220.05V I:0.0686A P:7.8859W PF:0.5226 Freq:50.00Hz Lamp Flux:776.887x1 lm		
NAME: Guide	TYPE: Indoor	WEIGHT:
SPEC.: 8W	DIM.:	SERIAL No.:
MFR.: kmutt	SUR.:	Shielding Angle:

Table--2 UNIT: cd

γ (DEG) \ C (DEG)	285	300	315	330	345														
0	206	205	206	205	206														
5	200	200	201	200	200														
10	195	196	195	196	196														
15	191	190	191	191	192														
20	186	185	187	187	188														
25	181	181	182	182	183														
30	177	177	178	178	177														
35	170	170	171	171	171														
40	162	162	162	163	164														
45	153	154	155	155	156														
50	146	146	147	147	147														
55	137	138	140	139	139														
60	130	130	131	131	132														
65	121	122	122	123	124														
70	111	112	113	113	114														
75	101	102	102	102	103														
80	75.2	76.4	76.1	74.2	76.3														
85	42.5	44.0	43.1	40.8	43.1														
90	15.0	15.6	15.1	12.1	15.5														
95	0.24	0.23	0.27	0.23	0.31														
100	0.24	0.30	0.27	0.30	0.24														
105	0.24	0.37	0.27	0.37	0.24														
110	0.24	0.40	0.27	0.44	0.24														
115	0.24	0.44	0.27	0.44	0.24														
120	0.24	0.50	0.27	0.47	0.24														
125	0.24	0.50	0.27	0.54	0.24														
130	0.24	0.50	0.27	0.50	0.24														
135	0.24	0.50	0.27	0.54	0.24														
140	0.24	0.50	0.27	0.60	0.27														
145	0.27	0.50	0.27	0.64	0.30														
150	0.37	0.50	0.30	0.74	0.37														
155	0.47	0.50	0.40	0.47	0.40														
160	0.47	0.50	0.47	0.47	0.44														
165	0.47	0.50	0.44	0.47	0.44														
170	0.47	0.50	0.44	0.47	0.44														
175	0.47	0.50	0.47	0.47	0.47														
180	0.50	0.50	0.47	0.47	0.50														

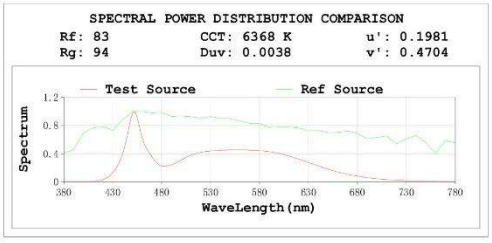
C Range: 0 - 360DEG
C Interval: 15.0DEG
Test Speed: HIGH
Temperature:26.8DEG
Operators:Kitivut
Test Date:06 June 2025

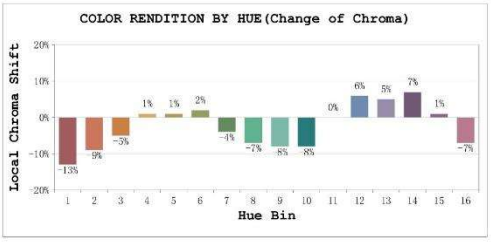
γ Range: 0 - 180DEG
γ Interval: 1.0DEG
Test System:EVERFINE GO-R5000_V2 SYSTEM V2.00.429
Humidity:42.9%
Test Distance:19.183m [K=1.0000]
Remarks: D3 with intensity Cal
%THDi=155.7437, %THDv=0.5246

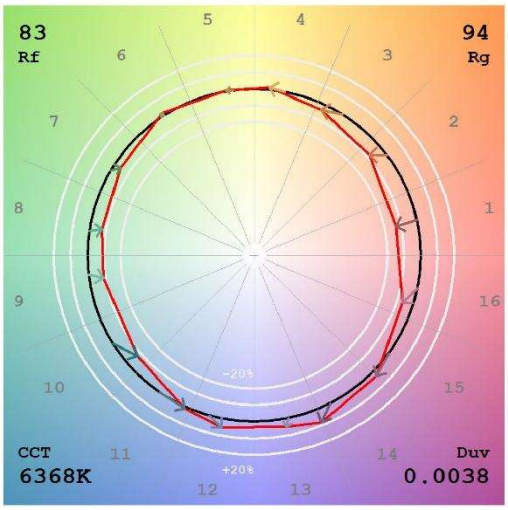
Spectrum Test Report ของหลอดไฟ A2

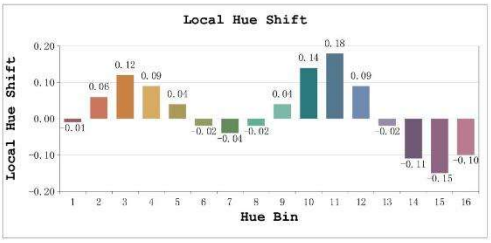
TM30

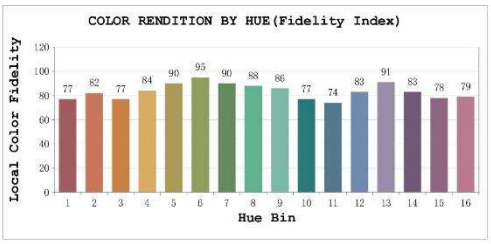
View Angle:2 Deg

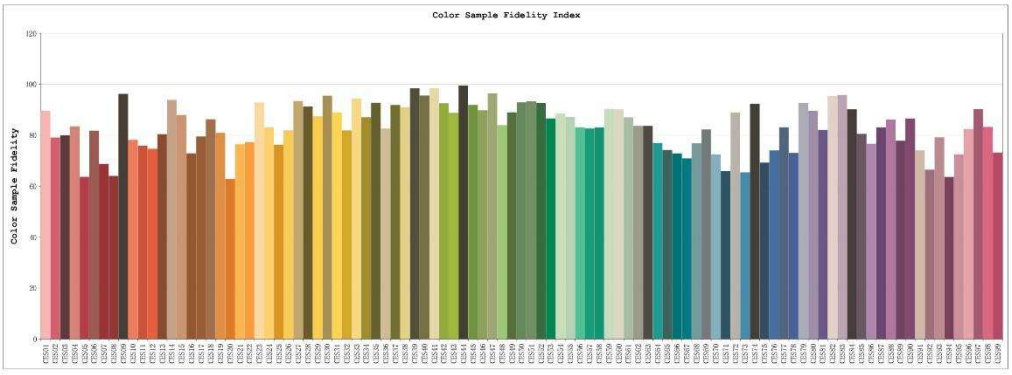












EVERFINE

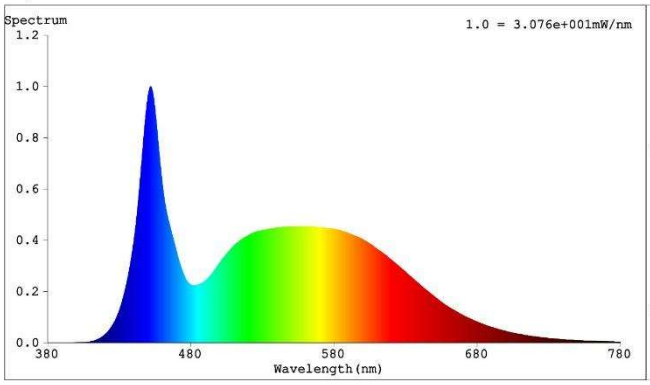
Spectrum Test Report

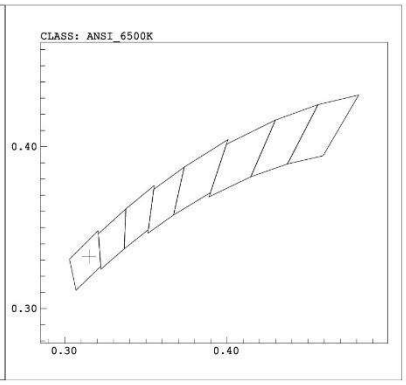
Sample	: Bulb 6500K	Date	: 2025-06-06 17:52:06
Specification	: 8W	Sam. Status	: %THDi=154.8192, %THDv=0.455
Sample No.	: Guide-1	Instrument	: HAAS-2000(EVERFINE)
Manufacturer	: kmutt	Test by	: Kiivut

Test Condition

Temperature	: 25.0Deg	RH	: 65.0%
WL Range	: 380nm-780nm	IP	: 49105 (75%)
Test Mode	: Accuracy Test	T	: 179 ms
		Sensitivity	: Low

Spectrum





Colorimetric Parameters

Chromaticity Coordinate: x = 0.3149 y = 0.3323 / u' = 0.1981 v' = 0.4704 (duv=3.80e-03) Du,Dv:-0.0030,0.0023

CCT= 6368K Prcp WL: Ld=491.0nm Purity=6.3%

Peak WL: Lp=452nm FWHM: =21.3nm Ratio:R=13.6% G=80.9% B=5.5%

Render Index: Ra = 83.3 TM30:Rf=83 Rg=94

R1 =81 R2 =88 R3 =91 R4 =83 R5 =82 R6 =82 R7 =89

R8 =70 R9 =9 R10=70 R11=82 R12=57 R13=83 R14=95 R15=77

WHITE:ANSI_6500K

Photometric & Radiometric Parameters

Flux = 913.28 lm Eff. : 114.12 lm/W Fe = 2.9352 W

Electrical parameters

V = 219.9 V I = 0.06960 A P = 8.003 W PF = 0.5228

Kdisp(IEC) = 1.000 Freq=50.00 Hz

Electrotechnology and Illumination Laboratory (E-LU)

http://www.ee.kmutt.ac.th

All Rf:

Rf1 = 89.5	Rf2 = 79.1	Rf3 = 79.9	Rf4 = 83.5
Rf5 = 63.6	Rf6 = 81.8	Rf7 = 68.7	Rf8 = 64.0
Rf9 = 96.2	Rf10 = 78.1	Rf11 = 75.9	Rf12 = 74.7
Rf13 = 80.4	Rf14 = 93.8	Rf15 = 88.0	Rf16 = 72.9
Rf17 = 79.5	Rf18 = 86.2	Rf19 = 80.9	Rf20 = 62.9
Rf21 = 76.4	Rf22 = 77.2	Rf23 = 92.9	Rf24 = 83.1
Rf25 = 76.2	Rf26 = 81.9	Rf27 = 93.4	Rf28 = 91.2
Rf29 = 87.3	Rf30 = 95.5	Rf31 = 88.9	Rf32 = 81.9
Rf33 = 94.3	Rf34 = 87.0	Rf35 = 92.7	Rf36 = 82.6
Rf37 = 91.8	Rf38 = 91.0	Rf39 = 98.4	Rf40 = 95.6
Rf41 = 98.5	Rf42 = 92.5	Rf43 = 88.8	Rf44 = 99.5
Rf45 = 91.9	Rf46 = 89.8	Rf47 = 96.4	Rf48 = 83.9
Rf49 = 89.0	Rf50 = 92.9	Rf51 = 93.3	Rf52 = 92.5
Rf53 = 86.5	Rf54 = 88.5	Rf55 = 87.2	Rf56 = 83.1
Rf57 = 82.6	Rf58 = 83.0	Rf59 = 90.3	Rf60 = 90.2
Rf61 = 87.0	Rf62 = 83.7	Rf63 = 83.7	Rf64 = 77.0
Rf65 = 74.2	Rf66 = 72.8	Rf67 = 70.9	Rf68 = 76.7
Rf69 = 82.2	Rf70 = 72.4	Rf71 = 66.0	Rf72 = 89.0
Rf73 = 65.4	Rf74 = 92.3	Rf75 = 69.2	Rf76 = 74.0
Rf77 = 83.1	Rf78 = 73.0	Rf79 = 92.6	Rf80 = 89.5
Rf81 = 82.0	Rf82 = 95.3	Rf83 = 95.8	Rf84 = 90.2
Rf85 = 80.5	Rf86 = 76.6	Rf87 = 83.1	Rf88 = 86.1
Rf89 = 77.8	Rf90 = 86.5	Rf91 = 74.0	Rf92 = 66.4
Rf93 = 79.2	Rf94 = 63.6	Rf95 = 72.5	Rf96 = 82.4
Rf97 = 90.2	Rf98 = 83.2	Rf99 = 73.2	